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Source: *Journal of Ecology*, Jun., 1988, Vol. 76, No. 2 (Jun., 1988), pp. 547-557

Published by: British Ecological Society

Stable URL: https://www.jstor.org/stable/2260612

Journal of Ecology (1988), **76**, 547–557

HOLOCENE GEOGRAPHIC SPREAD AND POPULATION EXPANSION OF *FAGUS GRANDIFOLIA* IN ONTARIO, CANADA

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SUMMARY

(1) Three radiocarbon-dated Holocene pollen sequences from southern Ontario, Canada, provide the basis for a reconstruction of changes in the distribution and abundance of *Fagus grandifolia* during the early Holocene.

(2) The three sequences are located on a transect from the area of present maximum abundance of *Fagus grandifolia* to its northern limit.

(3) At each site, *Fagus grandifolia* pollen concentrations/accumulation rates increase exponentially with time. The increases begin at least as early as the pollen is first detectable in the fossil record.

(4) 'Migration' or 'spread' of *Fagus grandifolia* across southern Ontario may have taken place at population densities too low to be visible in the pollen record.

(5) Low rates of increase of *Fagus grandifolia* populations during the Holocene appear to have been more significant in limiting its abundance and distribution than seed dispersal.

INTRODUCTION

The timing and rate of movement of tree populations during the Holocene has long fascinated biogeographers and ecologists, especially following the presentation of Davis (1976). An important attribute of the distribution of a taxon is its abundance, and the way it varies across the total range of the taxon. Abundance changes with time, like distribution, but the limitations of pollen analysis mean that it may not always be possible to distinguish the two (Bennett 1986).

Several authors have attempted to use changes in rates of increase of pollen frequencies as a basis for recognizing first arrival (i.e. as a measure of distribution change). Von Post (quoted by Jessen 1949, p. 104) defined 'the "rational border" of a pollen species as the level above which its frequencies rise markedly while the "empirical border" of a pollen species lies at the level above which it is continuously present with a frequency of not less than 1%'. These 'rational' and 'empirical' limits have been frequently used in discussion of pollen zone boundaries (e.g. Hibbert, Switsur & West 1971; Smith & Pilcher 1973), and of Holocene tree distribution changes (e.g. Chambers & Price 1985; Boyd & Dickson 1986). Davis (1976, 1981, 1983) has used sudden increases in pollen accumulation rates as evidence for arrival, based on comparisons of fossil pollen accumulation rates with modern estimates on either side of the distribution limits of a taxon (Davis, Brubaker & Webb 1973). Jacobson (1979) uses a 'major increase' in pollen of *Pinus strobus* as a basis for calculation of arrival times and migration rates. These approaches are similar in

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proposing that arrival of a taxon in a region can be detected from a change in the rate of increase of the rising pollen curve.

This approach has not found favour with all researchers. Faegri & Iversen (1950, p. 93) commented 'Previously it was assumed that when the curve rose above a few percent (the so-called rational limit) there was fair reason to think that the species grew within the area. In forested areas this may be correct in many, perhaps most, cases, but in this question dogmatism is particularly dangerous'. They discussed the importance of considering which other taxa are present in the pollen spectra and relative pollen productivities, and concluded 'It is hardly possible, by means of pollen analysis alone, to date the immigration of a forest tree exactly' (Faegri & Iversen 1950, p. 94). Tallantire (1972) pointed out that the empirical limit is related to the size of the pollen count.

Watts (1973) considered what actually happens at the margins of distribution of a tree when its range is increasing, and recommended that the whole of an increasing pollen curve should be compared with the classical models of population growth, which would imply that the curve cannot be separated into the 'long-distance pollen transport' and 'taxon present' components, separated by arrival times. Similarly, Godwin (1975) argued that, in using the rational/empirical limit terminology as an index of the spread of a tree taxon, it is assumed that the tree is spreading as a well-defined, steep front, whereas in fact 'every margin of distribution is diffuse and irregular with outliers, islands in a sea of different composition' (Godwin 1975, p. 489). Because of this, expansion can occur rapidly from established nuclei if climate changes. He concludes (p. 489): 'It is much safer to regard the successive empiric and rational "limits" as merely shewing relative increase in the taxon concerned, accepting that we do not here have the means of proving "immigration", but only of phases of expansion in frequencies of varying degree'. Following the suggestion of Watts (1973), Tsukada (1980, 1981, 1982a, b, c, d, 1983), Tsukada & Sugita (1982) and Bennett (1983) have shown that pollen frequencies do rise exponentially, as would be expected for an increasing population. The importance of geometric increase of numbers in populations has been emphasized in some recent models of spreading populations (Auld & Coote 1980), species abundance distributions (Hughes 1984, 1986) and the outcome of competition (Dorschner *et al.* 1987). Brubaker (1986) discusses Holocene changes in tree distribution and abundance from the standpoint of population dynamics and biological characteristics of trees, including population growth rate.

If increasing pollen frequencies do represent exponentially increasing populations, then the curves cannot necessarily be broken down into 'pre-arrival' and 'post-arrival' phases, and use cannot be made of 'sudden increases' or 'rational limits' as a basis for measuring tree spread, although they may give '*terminus ante quem*' arrival times. Thus, pollen curves at any one site would represent changing local abundances, and may not always be used as a measure of rate of spread to the pollen catchment of the site.

Clearly, plant abundances, as seen in the pollen record, cannot increase exponentially for ever, and logistic models have been used to allow for the declining rate of increase as populations increase (e.g. Tsukada 1981; Bennett 1983). The estimated values of r are not significantly different between exponential and logistic models (Bennett 1983), and the logistic model involves many assumptions (Pielou 1977) which are not easy to justify. In this paper an exponential model is used at three sites to examine the spread of *Fagus grandifolia* across southern Ontario, with regard to rates of change of abundance as well as rates of geographic spread of populations across the region. This is an approach which has not been adopted before and—in its basic assumption that changes in pollen abundances in time and space are reflecting population changes produced by time and the reproductive characteristics of tree species—it conflicts with arguments that such changes are produced by the response of tree populations to climatic change (e.g. Webb 1986). It may be difficult to resolve these lines of argument, but both are worthy of consideration.

Vascular plant nomenclature follows Scoggan (1978). All radiocarbon ages are given as uncorrected radiocarbon years before present (A.D. 1950) (B.P.). Calibration to calendar years, which would be more satisfactory for comparison with modern processes, is not yet realistically possible.

DATA

The data analysed here are from three Holocene lake sediment sequences in southern Ontario (Hams Lake, Nutt Lake, Lac Bastien; see Fig. 1). The three sites were chosen so as to be as similar to each other as possible, and to be as evenly spaced as possible from near Lake Erie, the area of maximum abundance of *Fagus grandifolia* during the Holocene, to its northern limit in Ontario (Fig. 1). Full details of the sites, pollen diagrams and dating are presented elsewhere (Bennett 1987). *Fagus grandifolia* was chosen for this study because: (i) it is the only representative of its genus in North America (Fowells 1965), so pollen of the genus is unambiguously referrable to a species; (ii) the relative frequency of its pollen in lake sediments is roughly equal to that of its occurrence in forests

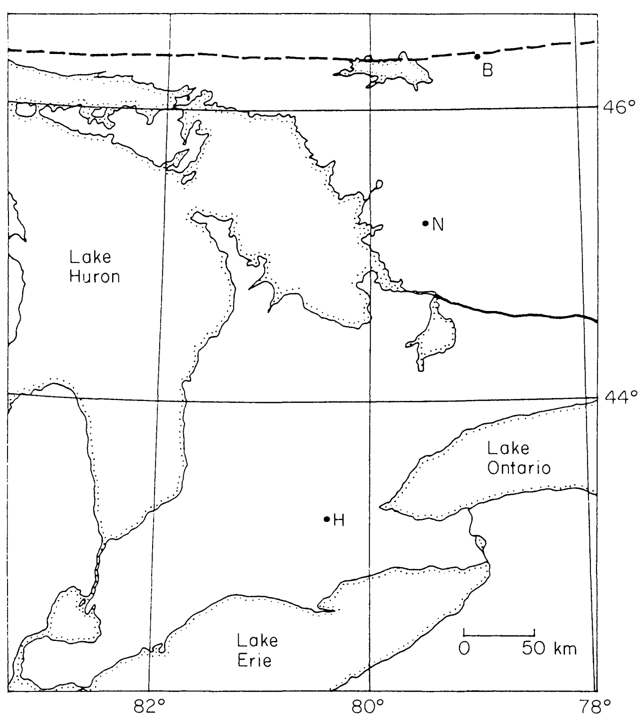

FIG. 1. Location map of sites in southern Ontario, Canada. Dots and letters indicate the position of the Holocene pollen sequences as follows: H = Hams Lake, N = Nutt Lake, B = Lac Bastien. A continuous line marks the southern edge of the Canadian Shield, and a dashed line indicates the approximate northern limit of *Fagus grandifolia.*

around the site (Davis & Goodlett 1960; Davis 1963; Livingstone 1968; Delcourt, Delcourt & Webb 1984); and (iii) its pollen is not widely dispersed, with few grains travelling more than 2·3 km (Bradshaw & Webb 1985). Long-distance pollen transport is not likely to be a source of error in this study because of these characteristics. It is assumed here that changes in frequencies of *Fagus grandifolia* pollen do track changes in the abundance of *Fagus grandifolia* within a short distance (*c.* 2 km) of each site. This cannot be demonstrated with certainty for these sites at the period of time in question, but seems a reasonable estimate. Long-term (> 100 y or about 5 cm sediment accumulation in these lakes) mixing of sediments could be a problem in the type of detailed analysis attempted here, but available evidence suggests that, for example, bioturbation is restricted to shallower depths (Davis 1974). No signs of sediment movement were detected.

Total pollen counts are about 1000 grains per sample for Holocene levels. Where the pollen count for *Fagus grandifolia* was less than ten grains (twenty-five grains at Lac Bastien), extra counts were made of *Fagus grandifolia* pollen and exotic pollen (added to determine pollen concentrations) until the *Fagus grandifolia* pollen count reached ten (twenty-five at Lac Bastien), to permit more precise calculation of *Fagus grandifolia* pollen concentrations than would otherwise have been possible. A higher total was chosen for Lac Bastien because a greater stratigraphic extent of the *Fagus grandifolia* curve had low counts.

Ten radiocarbon dates were obtained from each of Hams Lake and Nutt Lake, and five from Lac Bastien. Each site has a linear age–depth relationship, with no evidence for sediment focusing or dating errors (Bennett 1987).

RESULTS

The increasing concentrations of *Fagus grandifolia* pollen at each site were analysed through least-squares linear regressions of the natural logarithm of *Fagus grandifolia* pollen concentrations on depth (using the computer program REGRESS, written by J.M. Line). Taking logarithms of pollen data has several advantages, and has been used to display pollen data (e.g. Ritchie 1984, 1985). First, it permits examination of changing values over a greater range than an arithmetic scale. This is significant because a lake catchment may include more than 10^6 trees, and it is obviously of interest to know as much as possible about what is happening over the full range of this magnitude, and not just the upper 10^2–10^3 possible on arithmetic scales (Bennett 1985). Second, if a linear relationship exists between ln(pollen abundances) and time, then the gradient of the slope is equivalent to r, the intrinsic rate of increase, of classical population growth studies (Bennett 1983; Birks & Gordon 1985), and demonstrates exponential increase. This is based on the simple assumption that rate of increase of population size (dN/dt) is proportional to N, the number present, with r as a constant, and N proportional to the abundance of pollen in the lake sediment. Third, if the increasing pollen curve is, even in part, affected by processes other than the pollen production of the locally expanding population of the taxon concerned, this should be apparent as a deviation from exponential increase. In particular, if the early 'tail' of the increasing curve is due to long-distance pollen, or re-worked pollen, this should result in pollen abundances greater than would be expected from the exponential model during the early stages of the increase. The presence of a taxon in the catchment during the early 'tail' of the increasing pollen curve would be confirmed by discovery of macrofossils, but 'the probabilities must be extremely low that macrofossils would be found in sediment dating from a time when the local

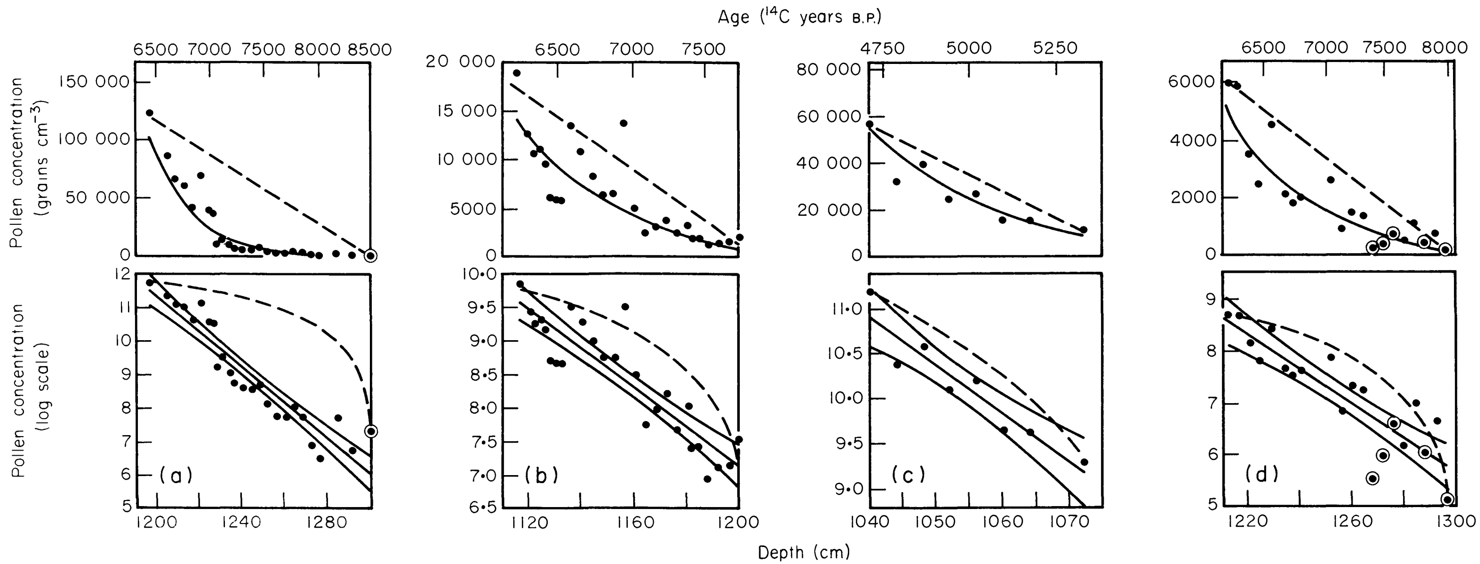

FIG. 2. Plots of *Fagus grandifolia* pollen concentrations, from sites in southern Ontario, Canada, and ln(pollen concentrations) against sediment depth and age, with best-fitting regression line and 95% confidence intervals for the line. Dashed lines are linear increases for comparison. Ringed points based on < 5 grains of *Fagus grandifolia* pollen. (a) Hams Lake, (b) Nutt Lake (first rise), (c) Nutt Lake (second rise), (d) Lac Bastien.

TABLE 1. *Fagus grandifolia* pollen abundance regression analyses

	Hams Lake	Nutt Lake I	Nutt Lake II	Lac Bastien
Number of levels	25	26	8	20
Correlation coefficient*†	−0·936	−0·898	−0·944	−0·876
Gradient*	−0·0526	−0·0291	−0·0535	−0·0322
Gradient‡	−0·00271	−0·00156	−0·00286	−0·00156
Doubling time (y)	256	444	242	444
Range§	220–305	369–559	180–373	350–608

* From regression of ln(pollen concentration) against depth.

† For all correlation coefficients, $P < 0{\cdot}001$.

‡ Regression gradient corrected for sediment age, using the sediment accumulation rates (Bennett 1987).

§ 95% confidence interval.

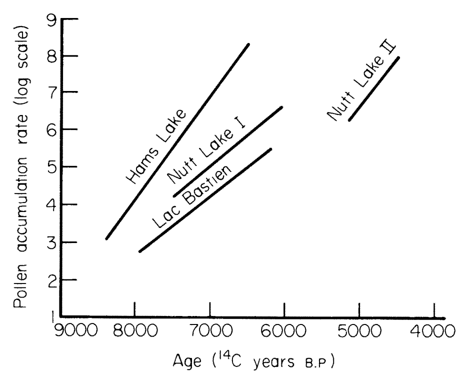

FIG. 3. ***Fagus grandifolia*** ln(pollen accumulation rates) against age: best-fitting regression lines from each site (Fig. 2, Table 1) compared directly. Data are from sites in southern Ontario, Canada.

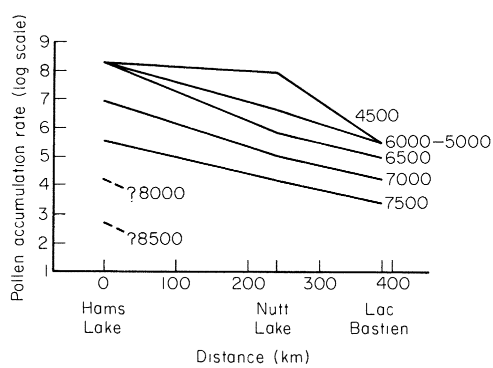

FIG. 4. ***Fagus grandifolia*** ln(pollen accumulation rates) against location on a south–north transect in southern Ontario, Canada, at 500-year intervals, in radiocarbon years B.P.

population of a taxon was so low' (Davis & Jacobson 1985, p. 365). Nevertheless, Davis & Jacobson (1985) did find *Picea* and *Populus* macrofossils from this time. Nothing can be concluded from the absence of macrofossils because of their general scarcity in lake sediments.

A disadvantage of plotting logarithms is that such plots emphasize low values, which are the least reliable. High pollen counts were made to overcome this as far as counting time constraints would permit. Also, even if tree populations were expanding exponentially, changes in climate or the arrival/extinction of other taxa during the increase would quite likely result in changes in the rates of increase. However, *Fagus grandifolia* is the last of the main tree taxa to become a significant component of the vegetation in southern Ontario during the Holocene and, on the independent evidence of climatic change from deep sea cores (e.g. Martinson *et al.* 1987) and ice cores (e.g. Dansgaard *et al.* 1982), climatic changes during the Holocene may be considered minor relative to the major change at the onset of the Holocene 10 000 years ago, and unlikely to have caused such important events as the spread and increase of *Fagus grandifolia* into southern Ontario.

The most useful measure of pollen abundance is pollen accumulation rate, plotted as a logarithm against sediment age. Here, because the age/depth relationship of all three sites is linear (allowing for the unavoidable uncertainties associated with all radiocarbon dates), and because overall sediment accumulation rates at the three sites are virtually identical (Bennett 1987), calculations have been made using *Fagus grandifolia* pollen concentrations and sediment depth, to avoid problems by adding errors of dating and sediment accumulation rates to both sides of the regression. Appropriate corrections were then made to the results for age rather than depth using the sediment accumulation rate constants (Bennett 1987). Doubling times (DT), in radiocarbon years, have been obtained from r, using the formula $DT = \ln 2/r$, as recommended by Harper (1977). Doubling times are used because they are intuitively easier to interpret than the raw r value (cf. use of 'half-life' for radioactive decay).

Plots of *Fagus grandifolia* pollen concentration and ln(pollen concentration) against sediment depth and age are given (Fig. 2) for the main, early Holocene increase at each site, plus a second increase at Nutt Lake following the mid-Holocene collapse of *Tsuga canadensis* pollen frequencies (see Bennett 1987). The best fitting least-squares regression line for ln(pollen concentration) against depth, together with a curve illustrating linear increase, is presented on each for comparison. Details of the regressions are given in Table 1. To facilitate comparisons between sites, the best-fitting regression lines from all sites are plotted against time (Fig. 3), and a profile across the range margin of *Fagus grandifolia* at 500 year time intervals (Fig. 4). Both Figs 3 and 4 use pollen accumulation rate values, essential for inter-site comparisons of absolute values.

DISCUSSION

In collecting the data set, much effort went into selecting similar sites and obtaining high pollen counts. It is difficult to see how the data could be improved without a vastly increased expenditure of time and effort, which might well fail because all such studies are ultimately dependent on the processes which created the sites more than 10 000 years ago. Interpretation of the data has to bear in mind the uncertainties of pollen analysis in general as well as any additional problems introduced by the methods used here. Only broad generalizations are possible, taking the data sets in their entirety: it would be foolhardy to try to interpret variations from level to level.

The four increases in abundances of *Fagus grandifolia* pollen examined here all appear to be exponential, although there are deviations from the regression lines (Table 1, Fig. 2), rather than linear or stepwise. The correlations of ln(pollen concentration) with sediment depth are all highly significant (Table 1). In this they support the results of Tsukada (1980, 1981, 1982a, b, c, d, 1983), Tsukada & Sugita (1982) and Bennett (1983) for other taxa in other areas. In the absence of major climatic or vegetational changes at the time of increase, it seems probable that the observed exponential increase does indeed represent exponentially increasing populations of *Fagus grandifolia*.

These analyses suggest that the beginnings of exponential increases are at least as early as it is possible, given counting time constraints, to detect pollen of *Fagus grandifolia*. Pollen frequencies increase from less than 0·05% of total pollen. *Fagus grandifolia* pollen frequencies at this level are already increasing in the same manner and at the same rate as at frequencies of 5% or more. This indicates that *Fagus grandifolia* is already present at each site (i.e. within 2–3 km of the lake) as early as it is possible to detect its pollen, and increasing. This conclusion is in full accordance with what might have been expected from population ecological considerations, since there is no reason to suppose that *Fagus grandifolia* populations would increase other than exponentially. (Hutchinson (1978) discusses special circumstances where an increasing population might not increase exponentially.) Moreover, *Fagus grandifolia* is monoecious (Hosie 1979), so one tree in the pollen catchment of the lake is all that is necessary to start a population increase (see Mollison 1986), at which level it could constitute only 0·0001% of the forest (Bennett 1985). Given that when populations increase they normally do so exponentially, and given the possible extreme low levels of abundance when invasion starts, the observed exponential increase should not cause surprise. However, it has not been noted before, and does have important implications for the interpretation of pollen curves. Before discussing these, some further points about the results presented here may be noted.

The rates of increase of *Fagus grandifolia* populations vary between sites, and even within a site (Nutt Lake) where local circumstances permitted a second increase (see Table 1). This may be related to latitudinal climatic trends (in southern Ontario): it was noted previously that *Fagus grandifolia* does appear to have increased most rapidly in areas such as extreme southern Ontario where it ultimately became most abundant (Bennett 1985, and unpublished). The faster increase of *Fagus grandifolia* at Nutt Lake after the *Tsuga* decline (Nutt Lake II) may indicate that the species composition of forest communities has some control over rates of increase. At the time of the first increase, 7500–6000 B.P., pollen spectra were dominated by pollen of *Pinus* (principally *P. strobus*), while during the later increase *Betula* (probably *B. lutea*) and *Acer* (mostly *A. saccharum*) were much more abundant (Bennett 1987). The increase of *Fagus grandifolia* following the *Tsuga* decline at Nutt Lake suggests that this event had considerable impact on the overall pattern of distribution/abundance of *Fagus grandifolia* in southern Ontario.

There is no evidence here that the 'intrinsic rate of increase' is a constant for *Fagus grandifolia*: it is clearly variable, probably being a function of a wide range of environmental and biotic variables. Although the two more northerly sites have lower rates of increase than Hams Lake, the reason for the low abundance of *Fagus grandifolia* at its northern limit is not a low potential for increase: it increased there just as rapidly as at Nutt Lake. The factors operating to determine overall abundance (or carrying capacity, K) are probably different from those controlling the intrinsic rate of increase (r). Abundance might, for example, be controlled by between- or within-species competition, whereas the potential for increase might be related more directly to climate and soils through flower and seed production on each tree.

There is no evidence that *Fagus grandifolia* spread across southern Ontario as a 'front' (see Fig. 4), at the spatial scale of Fig. 1. First, a more or less constant gradient of decreasing abundance with latitude appears to have been maintained throughout the period of increase during the early Holocene. Because at each site the increase appears to be due to local expansion, it would appear that the whole observed 'spread' of *Fagus grandifolia* is not a spread at all, but the gradual build-up of the population, limited by the locally possible intrinsic rate of increase. It appears later at Nutt Lake and Lac Bastien because of a slower intrinsic rate of increase than at Hams Lake and not because of a later arrival time (Fig. 3). Second, the populations at Lac Bastien begin increasing exponentially while those to the south (Hams Lake) are still at low abundances (Fig. 3). Population increase is occurring simultaneously at all sites, not later in the north as would have been expected if *Fagus grandifolia* spread as a 'front'. The observed pattern is one of exponential increase at all sites: 'spread' must be taking place at population densities too low to be observed.

Organic sedimentation at the three sites, marking the beginning of post-glacial environmental conditions, is time-transgressive, beginning at 10 400 B.P. (Hams Lake), 9900 B.P. (Nutt Lake) and 9500 B.P. (Lac Bastien). If the increasing gradients of Fig. 3 are extrapolated back to these times for each site, *Fagus grandifolia* pollen accumulation rates of 0·08 grains cm^{-2} y^{-1} (Hams Lake) and 1 grain cm^{-2} y^{-1} (Nutt Lake and Lac Bastien) are obtained. With a catchment area of about 2 km^2 (a crude but probably not unreasonable estimate) total pollen accumulation rates of 15 000 grains cm^{-2} y^{-1} for a fully forested pollen spectrum, and about 1000 trees ha^{-1}, these figures are equivalent to about one and thirteen trees, respectively, per catchment. In other words, the observed increases of *Fagus grandifolia* at each site could be explained by a small number of trees in each catchment at the very beginning of the Holocene. Equally, there could have been more trees as the basis for the increase, appearing later but still less than about 200 trees per catchment (the approximate number that would have been present when the observed increases began if the assumptions above are reasonable). If low numbers of *Fagus grandifolia* were present so far north so early in the Holocene, this would suggest that dispersal of *Fagus grandifolia* seeds is much wider than has hitherto been appreciated. It does appear that, in its Holocene history, *Fagus grandifolia* is limited more by the intrinsic rate of population increase than by dispersal capability. In contrast, Ritchie & MacDonald (1986) have demonstrated that *Picea glauca* may be limited by seed dispersal characteristics, with spread taking place most rapidly in circumstances especially favourable to seed dispersal. The form of the increasing pollen curves for *Picea glauca* in regions of slow and fast rates of spread are not noticeably different (Ritchie & MacDonald 1986, Figs 4, 5 and 7), which suggests that, as with *Fagus grandifolia*, 'rate of spread' and 'rate of increase' are distinct attributes, subject to different controls and limitations.

The results discussed here suggest that the Watts (1973) and Godwin (1975) model of tree population expansion is a more appropriate description of what actually takes place than any model which seeks to split the increasing pollen curve into two or more phases. Forest trees are organisms whose potential for increase in number is limited by intrinsic biological properties. No climatic or environmental conclusion should be based on 'appearance' of any taxon in the pollen record without due consideration of this (cf.

Dexter, Banks & Webb 1987). A small population could increase exponentially for long periods of time, and be palynologically silent, before the effects became dramatic (see Fig. 2a): anyone who has watched a bank balance grow by compound interest will be aware of this. 'Intrinsic rate of population increase' may be added to 'rate of spread' and 'distance from refugia' as a biological mechanism explaining the sequence of forest tree 'arrivals' observed in many Holocene pollen diagrams.

ACKNOWLEDGMENTS

I thank H. J. B. Birks, B. Huntley, H. F. Lamb, G. M. MacDonald and J. C. Ritchie for critical comments on this work, S. M. Peglar for drawing the diagrams, the Natural Sciences and Engineering Research Council of Canada for a postdoctoral fellowship, and J. C. Ritichie for his support.

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(*Received 1 June 1987*)